
The Key Concepts of Territorial Analysis: from the Actors to the Entrepreneurial Ecosystem

The slowdown in economic growth and the implementation of budgetary control policies contributed to question the major national industrial programs for the benefit of more decentralized initiatives and entrepreneurship. At the same time, the creation of large markets (including the European Union, North Atlantic Free Trade Association (NAFTA), and the Association of Southeast Asian Nations (ASEAN)) favored the implementation of policies which were more closely aligned to the targeted territory and economic players. In a more geographically open environment with greater budgetary constraints, the development and competitiveness of territories became objectives for public decision-makers and the subject of study for economists.

From the development of nations according to [POR 98], the stakes moved towards the so-called “winner regions”¹, to paraphrase the work of Benko and Lipietz [BEN 92]. An analytical framework was developed, as were models which aimed to break down the mechanisms by which some territories have succeeded in expanding and making progress while others seemed to encounter more

¹ Translated from the French title “Les régions qui gagnent”.

difficulties [BEG 02, ASH 11, BOS 04]. Previously acknowledged as a key factor in economic dynamics through the theorization of the spillover effects of entrepreneurship [AUD 95], innovation was placed at the center of these approaches. From innovative milieus [AYA 86, CAM 06] to *smart specialization* [FOR 15] via industrial districts [DEI 94], learning regions [MOR 97] or *innovation clusters* [SAX 96], the majority of approaches sought to evidence the combination of business, territory and innovation which produced competitive territories.

The notion of entrepreneurial ecosystems, which emerged at the beginning of the 21st Century, drew upon the source of these analyses of territories, although distinguished by the attention that it pays to the links between business and territory, and the intervention of other actors within the system, more than their respective characteristics. This systemic nature is indeed specific to it, as Laperche and Uzunidis underline [LAP 11, p. 167]. According to these authors, “the territorial economy has specific systemic characteristics, which are, nowadays, the reason for numerous analyses upon its re-construction from the implementation of different types of innovation processes”. A territory is, therefore, not a simple static receptacle or the substratum of production activity in which businesses might draw the necessary resources following a unidirectional plan. On the contrary, as per an evolutionary point of view, relationships between firms and territories are bidirectional. The ecosystem conditions the activity of the players comprising it, who in turn can influence ecosystem operation through their strategies and the interactions they have with other players who comprise the ecosystem. While doing so, entrepreneurial activity contributes to building, modifying and restructuring the entrepreneurial ecosystem from which it originates.

To analyze the internal operation of entrepreneurial ecosystems and to show the overall architecture of it, this chapter bridges the two fields of economic analysis which generally remain separate. On the one hand, we use works from industrial economics, and more

particularly the economics of the firm, to pinpoint the determinants of growth. Size, age, activity sector, forms of governance and financial and industrial strategy are then presented as the key explanatory factors for growth, employment, turnover or profit. On the other hand, we draw from the field of economic geography which emphasizes the explanatory factors of territorial performance: location and agglomeration economies (diversification or specialization), human capital, innovation, etc. These two spheres of economics have allowed for both the creation and dissemination of knowledge of business strategy, and of local policies.

Studies carried out in the field of economic geography have also highlighted the relationship between territory and firms. In the majority of cases, this includes the affirmation that it is necessary to attract firms, favor the creation of new entities and ensure that they are protected from failure, as territorial performance ultimately depends on the economic well-being of the respective firms that it accommodates.

The works inspired by or coming from within the field of the so-called “new” economic geography have gone further in demonstrating the influence of territory upon actors’ behavior, by identifying types of socio-economic fabric which are more or less capable of influencing economic efficiency. It is a matter, for example, of geographical proximity, ranges of activity, factor endowments and agglomeration processes concerning those activities which are either similar or diversified. These relationships are essentially used to explain the changes in production systems linked to the start-up and winding-down of businesses. The rise in promotional policies within territories and territorial marketing attests to the impact which these research works have caused. They contribute to the development of horizontal policies helping all firms in the same way, or vertical policies which place stress upon sectoral dimensions.

Although it has become commonplace to consider that by their conglomerations, firms make up territorial wealth, we know less of how the territory can influence the evolution of the firm it allows in,

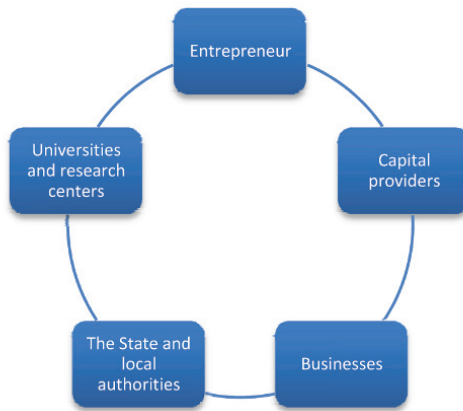
and in particular, how the actions of firm affect the territory. There remains much to do in order to understand the one-to-one linkages between these two levels. Research studies which allow us to consider the firm and the territory in the same context are rare. Such is the objective which the notion of the entrepreneurial territory responds to, its objective being to clarify this question by the fusion of several considerations:

- the entrepreneurial context and the business climate (rate of firm creation and rate of business sustainability up to one and up to five years, the share of small and medium enterprises (SMEs) in relation to all of the businesses located in the territory, etc.);
- the institutional context favoring entrepreneurship (policies and structures accompanying business creation, tax incentives, the existence of business incubators, graduate training schemes, etc.);
- amenities (environmental context, pro-cultural policies, housing, living conditions, etc.);
- the technical and scientific contexts (the existence of a university center, well-known business or engineering *grandes écoles*² and research centers etc.).

These fundamental components are necessary, but, alone, are not sufficient for the operation of an innovative entrepreneurial ecosystem.

The regional entrepreneurship acceleration program (REAP) set up by the Massachusetts Institute of Technology (MIT) is an example of the desire to anchor firms to the territory and make the two components of the economic system develop in tandem. It is based upon the following diagram (Figure 2.1) which places emphasis on both the coexistence of different actors and their function within a system, oriented towards innovative entrepreneurship and economic development caused by grouping together economic actors.

2 These are highly competitive French higher education institutions.



Source: <http://reap.mit.edu>

Figure 2.1. *The MIT Regional Entrepreneurship Acceleration Program*

At this stage, the question that players ask and which this chapter provides answers to is the link between entrepreneurial capacity and the local economic system. To achieve this, it is imperative that we return to the economic players [2.1] before describing the operation of the system [2.2] and strategies allowing for its activation.

2.1. The actors and components within the entrepreneurial ecosystem

The terms territory and entrepreneurial ecosystem are part of the concept which has been disseminated most widely throughout the public domain. It creates the territorial “pact” for growth, employment and innovation of the Assembly of French Local Authorities³, as the *Institut de l’Entreprise*⁴ has observed⁵. There is a general consensus as to the definition of the entrepreneurial ecosystem: according to the definition given by Moore [MOO 93], it is an economic community

3 Available at: <http://www.adcf.org/files/Pacte-economies-territoriales-25e-convention-AdCF-octobre2014-web.pdf>.

4 Institute of Business.

5 Available at: <http://www.institut-entreprise.fr/les-publications/entreprises-et-territoires-pour-en-finir-avec-lignorance-mutuelle>.

supported by the interaction of individuals and organizations (Box 2.1).

The entrepreneurial (or business) ecosystem is characterized by the diversity of players which comprise it and through their shared objectives and skills. It is a system of heterogeneous independent players who, in so-called “coopetition”, establish a common strategic interest by sharing resources, skills and norms/standards of strategic behavior which take many forms (technological standards, shared know-how, shared visions, informal rules, etc.). We owe both the expression and the notion of the entrepreneurial ecosystem to Prahalad [PRA 05]. This specific of ecosystem allows participating actors (that is to say the individuals, businesses and society as a whole) to effectively merge resources with the aim of creating economic wealth and prosperity. It quickly gains a political significance and is presented as a catalyst for economic progress within stable economies and a driving force for declining economies [RAM 12] or indeed for economies in transition.

CLUSTER

In his seminal work, Michael Porter [POR 98] (see Chapter 1) defines a cluster as a group of decision-makers who devise and maintain a relational system allowing them to increase their business and growth opportunities: “it takes shape as a geographical cluster of businesses, linked by specialized suppliers, service providers, related industrial firms and associated institutions (for example, universities, standardization agencies, or professional organizations) in a particular sphere which both confront and cooperate with each other”. By adding the spatial element to his so-called “diamond”, Michael Porter largely contributed to popularizing the concept that territory is not a neutral space but, on the contrary, it is a melting pot for competitiveness.

Businesses within the cluster benefit by a local environment of competing firms, specialized resources and institutions, which encourages knowledge transfers and stimulates their competitiveness. Thus, the cluster is considered as a system which is “strengthened from within” and which is wealth-producing, in an endogeneous manner. The four elements of the so-called competitive diamond (conditions for growth, firm strategies and competition, conditions related to demand and favorable related industries) gathered in the same cluster, has a triple beneficial effect, by concurrently stimulating

productivity, innovation and entrepreneurship. The grouping of a critical mass of resources and skills upon the same site within the same sector provides production gains for businesses located within such territories and procures a comparative advantage in global economic competition for a territory (whether an individual region or nation).

The success of the cluster notion comes from its capacity to use different approaches and to find a place within different disciplinary fields. The spatial economy conceives it as an agglomeration space, the regional, and industrial economy define it as a space involving industrial relationships between collective entities, whilst socioeconomics sees it as a space involving relationships and social integration.

The Ecosystem

The “ecosystem” concept was popularized by the link which Christian Blanc made upon the development of growth ecosystems. However, this does not provide a sufficient definition of it. The notion of an ecosystem is a biological metaphor for which Moore [MOO 93] was its pioneer when studying the business ecosystem for large corporates. He describes the ecosystem as a bipartite organization made up of a central core and a periphery in which various actors play a part: “a business ecosystem is made up of customers, market intermediaries (...) suppliers, and, of course, oneself. These might be thought of as the primary species of the ecosystem. Yet, a business ecosystem also includes the owners and other stakeholders of these primary species, as well as powerful species who may be relevant in a given situation, including government agencies and regulators, and associations and standards bodies representing customers or suppliers”.

At the root of the different types of ecosystem identified by Koenig we also find “an economic community based upon interaction between businesses and individual economic actors” [MOO 93]. The entrepreneurial or business ecosystem is characterized by the diversity of actors which comprise it, through their common objectives, and skills which are then shared. You can define it as a heterogeneous system of independent actors in so-called “coopetition”, who set up a community of strategic interests, by sharing resources, skills norms/standards of strategic behavior. This may take many forms (technological standards, shared know-how, visions, informal rules, etc.).

For further information, see Koenig’s work [KOE 12].

Box 2.1. *From the cluster to the entrepreneurial ecosystem*

If the viewpoints upon the definition of the entrepreneurial ecosystem are somewhat convergent, those about the nature of actors who comprise this entrepreneurial ecosystem are, on the other hand, different, or indeed divergent. The heterogeneousness of the actors comprising the ecosystem brings those specializing in the entrepreneurial ecosystem closer together. However, whilst some even emphasize the actors, others involve both the formal and informal relations which they sustain.

The least stringent definition is that which Moore [MOO 06] provides. For him, the business ecosystem and its sub-ecosystems comprise communities of economic actors whose individual entrepreneurial activities have an impact on the entire community. Clusters of customers, suppliers, producers, financiers, professional associations, public bodies and other stakeholders are involved and interact to produce goods or services. From this general framework, the analyses changes in a more normative direction, seeking to identify the factors which are necessary to form entrepreneurial ecosystems.

Cohen [COH 06] identifies seven of them: the informal network, the formal network, the university, the government, professional material and support, capital services, and talent. Four years later, in 2010, Daniel Isenberg, in an article in the *Harvard Business Review*, analyzed different examples of entrepreneurial ecosystems throughout the world (Puerto Rico, China, Iceland, India, etc.) and showed the thirteen factors which are necessary for their existence and operation: leadership, government policy, culture, success stories, human capital, financial capital, entrepreneurial organizations, education, infrastructure, economic clusters, networks, support services and main customers. He concluded that entrepreneurs succeed better when they have access to financial, human and professional resources which they need and when they operate within environments where entrepreneurship is favored and is protected by the implementation of public policies in its favor⁶.

⁶ The gap between this concept of the entrepreneur, slotted into a system upon which his success largely depends, is quite astonishing, as is that taken by the Schumpeterian or Austrian approach, which makes the entrepreneur a hero or a leader.

Owing to the work of Suresh and Ramraj, this drift towards the relationship to the detriment of the player now goes one step further [RAM 12]. Starting from a hundred case studies from in-depth fieldwork in the state of Tamil Nadu, in southern India, they draw up a conceptual model of the entrepreneurial ecosystem. This rests upon eight support systems which come together to form a unique framework. This model has the following elements [RAM 12]:

- the term moral support refers to the role played by both the family and the businessman's support circle;
- financial support which comes from many partners with varying levels of commitment, includes family, banks, venture capitalists, friends, teaching establishments business angels and other minor investors from capital markets, financial and credit institutions and suppliers, etc;
- the support network integrates all of the entrepreneurship support organizations. This could include, for example, particular industrial organizations, former student associations, online networking sites and groups of suppliers and distributors;
- government support which is apparent through the implementation of clusters, incentive systems, business incubators and infrastructure;
- technological support which is provided by centers financed by so-called business incubation governments, new technologies developed within educational institutions and the business incubators themselves;
- market support refers to all business opportunities resulting in sales and new contacts with potential customers;
- social support which is linked to prizes and bonuses awarded by associations and institutions but also to both the tolerance of failure and to the positive view that society has of business;
- environmental support includes all of the natural resources and locally available amenities.

These components must be located close to each other to operate efficiently. External agglomeration economies (Box 2.2) are thus at the very foundation of entrepreneurial ecosystems. The works in this field identify two types of agglomeration economy: localization economies and urbanization economies. These external effects may be described according to the mechanisms which underpin them. Localization economies may result either from the size or the number of businesses set up within a territory. Extensive use of a local factor may also promote external savings within a group of businesses belonging to the same sector [DUR 01]. According to Catin [CAT 97], urbanization economies are both external to the firm and external to the firm's particular industry. For the whole of the industry, these economies result in particular from the economic size of the agglomeration, the population concentration (density) and the availability of infrastructure and services for business. The reduction in production costs, induced by agglomeration economies, makes performance gains possible, which, in turn, may favor job creation within firms which benefit from it.

The works of Alfred Marshall [MAR 90] are behind the concept of agglomeration (see section 1) which constitutes an economic advantage, as it allows for the cost reduction of moving goods, people or ideas. From there flows the notion of external agglomeration economies which refers to the over-performance of businesses located in a dense or clusterized economic environment. These advantages originate from three sources. The first two are economies linked to the proximity of a large number of specialized suppliers (of intermediate goods and services) and those linked to the presence of a larger specialized and stable workforce, which was resumed in the "new" commerce theories and the new "economic geography" [KRU 91]. Thirdly there are external economies which are linked to the dissemination of knowledge (technological externalities) introduced within endogenous growth models which integrate the stock of knowledge or human capital as a means of explaining regional growth.

The issue is then knowing why individuals or businesses tend to group together. The supporters of the neoclassical models of perfect competition forward the first tier of reasoning according to which economic actors wish to be clustered together having realized the advantages of proximity.

The desire for agglomeration also comes from a desire for diversity evidenced by consumers while each firm tends to specialize in the production of a good where it produces yields on an increasing scale. Both firms and consumers find conglomerating to be advantageous, as the more the density increases, the more the number of firms grows and the greater the range of products, thereby increasing the differentiation.

Finally, it is in businesses' interests to be as close as possible to each other (the principle of minimal differentiation) to attract the highest number of consumers as possible. Models such as Hotelling's straight line (see section 1.1.5) show this, by taking account of spatial competition with strategic interactions between businesses.

One particular author has played a key role in this revival of spatial economics. He is Masahisa Fujita, a Japanese economist, who, as well as Paul Krugman, with whom he frequently worked, laid the foundations for the new economic geography by showing that external savings are derived from the interactions between economies of scale at firm level and transport costs [KRU 12, pp. 9–10].

Box 2.2. *Agglomeration economies*

The synthesis of different works devoted to entrepreneurial ecosystems allows for the description of various basic components and their organization. However, despite being locally-based, this approach is not allied to regional or local concepts. The objective of the organization of an entrepreneurial ecosystem lies in the strengthening of both bottom-up development and local skills. In order to achieve this, the alignment of various components is necessary to establish synergism. Proximity or location within the same territory of basic resources necessary for the ecosystem is no guarantee of the operation of the latter. On the contrary, as different authors who are in agreement within various schools of thought have emphasized, more so are proximity or indeed the clustering of variables. The need for activation through external means differentiates the proponents of a new economic geography, for whom agglomeration is enough to generate productivity gains, from authors' reasoning in terms of external links between the different actors located within the same territory.

Figure 2.2 shows this. The coexistence of building blocks (production factors, spatial and organizational aspects, human and social characteristics) requires appropriate policies to boost overall territorial performance. Data and economic intelligence are thus used for common aims. Far from being infrastructure builders, or even production system operators, in this case local authorities become facilitators of economic activity responsible for territorial coordination by the creation through links between actors on the ground, some of whom may be behind the plant set-up.

The potential of the coexistence of these various factors is maximized by public action producing both information and economic intelligence. When applied to economic policies, this approach consists of the production of knowledge serving the economic aims of an organization, which in this case is the territory. In France, its promotion dates back to the report *Economic Intelligence, Competitiveness and Social Cohesion* which was delivered to the Prime Minister in 2003⁷. In section 2.6 of the report, titled “Economic intelligence and territory”, the reporter Bernard Crayon views it as a policy of competitiveness, economic security and influence, particularly as compared to organizations and also for training purposes. It is based on a public policy for identification of sectors and strategic technologies, organization and convergence of interests between the public and private spheres. Territorial marketing provides a “kit”, as a result of the transposition of management tools for big business and ready-made techniques, which is implemented by specialist practices or agencies.

In-depth analysis of the three horizontal blocks (base resources, economic intelligence and performance or value creation) is simplified in Figure 2.1, which should be read from the bottom-up. It recalls that the entrepreneurial ecosystem needs basic resources from which to both draw from and “feed” from to ensure its continued operation. These are represented by the lower stratum, or sub-stratum, which brings together two families of elements:

⁷ It was published by *La Documentation française* (a French publisher) in 2015.

– the superstructures which bring together organs of a political and legal nature, that is to say institutions representing both the state and local authorities, as well as academic and collective representatives. Examples include culture, finance networks and the demand emanating from businesses and households;

– intangible assets generate social links. Amongst these are actor networks, financial resources, support services and invisible factors behind local over-performance, as Carré and Levratto [LEV 13a] have shown. It is a matter of knowledge, talent and coordinating territorial initiatives and projects owing to the emergence of a leadership which not only allows players to know their milieu but also, and especially, to act consistently.

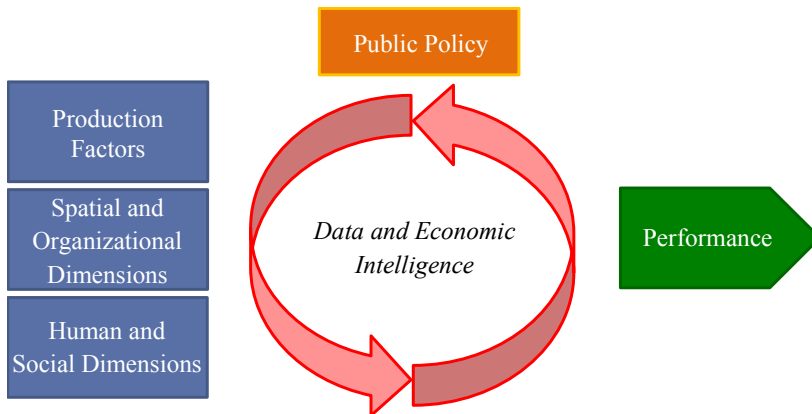


Figure 2.2. *Territorial strategic activation policies*

Entrepreneurial activity is anchored within this economic substratum. On this point, works in the field of entrepreneurial ecosystems re-examine the notion of the entrepreneur, returning to a Schumpeterian concept which favors the entrepreneur's role as an innovator. Indeed, however much the entrepreneur is innovative and epitomized, he is no less integrated into a group or networks, combining strong, formal or weak links. It is from the latter that creativity and new ideas emerge.

Young innovative firms and high growth businesses, or “gazelles”, are therefore at the forefront of analyses on the creation and operation of entrepreneurial ecosystems. However, they cannot operate in an autonomous and remote way, since the diversity of the production fabric is both a source of resilience in the event of economic crisis, and of performance, for those businesses which comprise it. The existence of a diversified local economic system according to the industries in which the companies operate, their sizes, ages and the relationships they maintain with other business entities is a crucial aspect to be considered. Another important, but still barely considered, point to observe is the financial linkages between the local plants on one hand and the heads of groups, as in the case of subsidiaries or commercial units for example, within franchise networks on the other. These entrepreneurial ecosystems are also based upon projects and joint objectives which are both developed and implemented. Having fulfilled these conditions, the issue is thus to determine which objectives to achieve. We can define these in terms of value creation whether the latter is exclusively commercial or equally if it includes non-commercial aspects, such as the promotion of new well-being indicators (Figure 2.3).

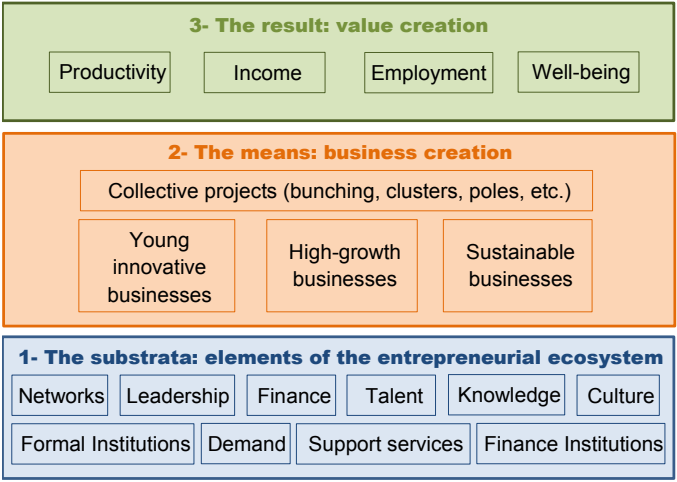


Figure 2.3. *The various strata comprising the entrepreneurial ecosystem*

Since the first works on localized production systems appeared, the issue of local business anchorage experienced both a significant development and a profound change (see Chapter 1). This two-pronged movement was analyzed in a review of research works on the start-up of new businesses at regional level published in four special editions of the academic review *Regional Studies* over a period of thirty years, completed by Fritsch and Storey [STO 14]. They stress the increased acknowledgment of the role of official institutions and intangible factors, such as social capital and the culture of entrepreneurship during the period. In the end, these approaches to entrepreneurial ecosystems differ from spatial analyses of the production systems which preceded them by taking into account the links between players which are either disregarded or summarily considered in previous studies. By giving priority to the links between players, they expand upon them while also qualifying them through studies which insist the agglomeration of firms as an essential or a minimal condition, to allow clusters to achieve production gains.

In fact, entrepreneurial ecosystems have placed emphasis on three factors which will serve as a guide to develop territorial public policies. The first is the presence of entrepreneurial activity and, particularly, a breeding ground for high-growth businesses. The second concerns the territorial environment and the conditions needed to produce ambitious Schumpeterian entrepreneurship. The third is the link between the previous two factors as the operation of the business ecosystem depends upon both framework conditions and the local environment. This novelty, in light of the previous approaches, favored the emergence of public policies which were different to those which, for a long time, founded local development schemes, changing from judgment-based assistance and individual support to businesses with a collective approach based upon a project rationale [LEV 09].

2.2. The operation of the entrepreneurial ecosystem

As is the case with all ecosystems, the entrepreneurial ecosystem requires a mix of players and linkages between them to operate in an

efficient and sustainable manner. Unlike natural ecosystems, the entrepreneurial ecosystem is built with the aim of achieving economic efficiency, which requires that its purposes are permanently defined. The primary objective of the driving force or forces within an entrepreneurial ecosystem is thus to develop synergy between the various participants and to ensure that they converge on a shared objective. In order to succeed, the challenge is then to identify the sources for local performance and to organize resources allowing its achievement. The diversity of forces at work demands activation of the linkages between each ecosystem component and the objective to achieve. Contemporary literature identifies three major categories of local performance determiners which are respectively related to macro-, meso- and micro-levels of economic analysis. They are summarized by Figure 2.4, taken directly from the works of Carré and Levratto [LEV 13a].

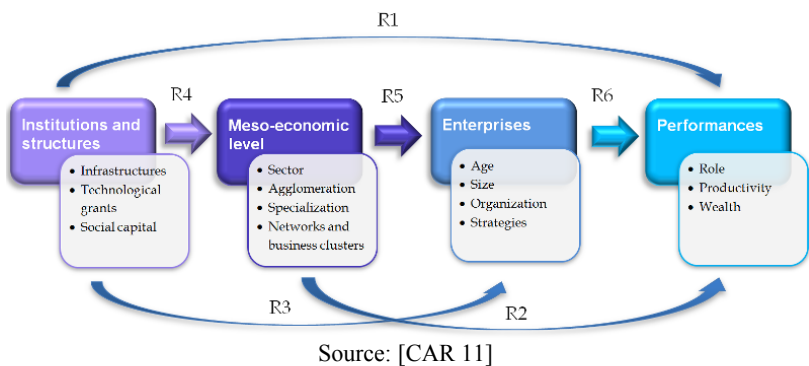


Figure 2.4. *Different levels of localized business performance determiners*

This evaluation grid takes account of the multiplicity and diversity of the basic elements of territorial development. Some, such as infrastructures or the creation of business networks, are controllable. Others, such as the age of firms or social capital donations, are either totally exogenous or inert. Faced with all of these heterogeneous

development factors, coherence is more important than juxtaposition. This approach leaves the way open for local public action directed towards efficient organization as a result of conditions that allow development per the territorial characteristics that will stimulate it. A dual flow made up of direct relationships (R1, R2, R6) and indirect relationships (R3, R4, R5) describes the relationships between the different territorial components and their performance. We will briefly recapitulate the material in what follows.

The relationship between, on the one hand, institutions and structures, and, on the other hand, the territory, serves as a reminder that infrastructural, institutional or social baseline conditions determine the evolution of territories. Alongside start-up resources, collective memory, institutions and the politico-legal framework are reasons for dependency regarding business trajectory. Meso-economic relationships supplement the effects of the broader context. Such relationships originate at the sector level or within businesses clusters concentrated in the same territory. This link is described in the R2 relationship, the ideas of which are closest to the current economic geography [KRU 91a, KRU 91b, STO 06]. This reiterates that the concentration of economic players provides advantages on account of positive externalities which are caused by the agglomeration of different firms. This rule is true for the Marshall-Arrow-Romer externalities which chart the growing relationship between interactions and the degree of firm specialization, as well as the externalities identified by Jacobs. These emphasize that interactions are all the more important as clustered firms are distinguished by a high degree of diversification of economic activity. The greater the concentration of economic players, the higher the increase in their respective economic performances. This is illustrated by the numerous works upon the metropolitan phenomenon. Indeed, Jean-Baptiste Say and Alfred Marshall have also evidenced it (see section 2.1).

The characteristics and strategies of businesses set up within the territory also contribute to determine its performance, as the R6 relationship emphasizes. This relationship between a firm and its

milieu depends on its characteristics, its strategy, its organization and its governance. Taking account of business behavior modifies the two previous relationships mentioned by reducing the impact of structure upon behavior. Taking the point further, this applies all the more so in view of the fact that the behaviors are not exclusively caused by the business structures.

The last three relationships illustrate the assumed effects of the “base conditions” upon productive “meso-systems” (R4) and upon business behavior (R3 and R5). Their presence underlines the impact of different determining factors on the performance of territories considered by analyzing specifics across the sectors. Some examples enable us to explain how this works.

The relationship (R4) between institutions and local performance may manifest itself by the way that national rules (e.g. competition law, insolvency law, the tax system and tax exemptions) influence the organization of sectors, agreements and business restructuring. The analysis of regional innovation systems equally provides substance to this relationship. Structures and organizations dedicated to research and innovation may be in line with economic activities.

The R3 and R5 which link the institutions and the meso-economy respectively to firm strategies complete the relationship (R2). They emphasize that economic players and businesses are dominated by the same principle which is caused by local or organizational structures of the milieux in which they operate. As such, this relationship runs counter to a direct link which is generally asserted between, on the one hand, individual behaviours, and on the other, organization of the local economy and territorial performance. Indeed, there is no reason why local firms concentrated within the same sector (the Marshall-Arrow-Rommel (MAR) hypothesis) or those dispersed within different activities (the Jacobs hypothesis) should make up a homogeneous whole from the point of view of both their capacity and their will to interact. On the contrary, there is every reason to consider that they can be distinguished, by having the capacities

and specific strategies which will have decisive consequences on their performance, but also, more generally, on the existence of the methods of interaction which shape territorial performance; likewise for the phenomenon of agglomeration. There is no basis for assuming that it will take a homogeneous structure and that the activation of potential connections between players will disregard both socio-economic profiles of territories and the strategies of the same players.

This concept serves as a blueprint for numerous studies into territorial economics. For example, it applies in the works on the microeconomic foundations of institutional inertia factors and upon the impact of institutional factors on adaptation behaviors [SER 03]. We find these self-same feedback loops and reinforcement effects in the analysis of the agglomeration processes producing agglomeration effects [BAU 97] which provides micro-tools for developmental inequalities across territories [KRU 91a, KRU 91b]. Interactions between local development determiners are also at the heart of a conventionalist approach initiated by Storper [STO 97] for whom baseline conditions contribute to shaping behaviors either directly or indirectly or to introducing routines which determine local specifics. Finally, Capello's interregional relationships [CAP 07], Casson's hierarchical relationships [CAS 01] and the problem of the local business within a global economy [SCH 04] provide additional contributions as to the variables to integrate into this local level analysis.

In France, competitiveness poles (Box 2.3) represent the type of territorial organization which is, in fact, closest to the entrepreneurial ecosystems. The structure of competitiveness poles differs from the approach of localized production systems (LPS) which came before them. Introduced at the end of 1990s, LPSs sought to institute proximity dynamics between businesses and their environment owing to an approach of contract-based links between a region, having the lead role, and the state, intervening in a support role. Competitiveness poles differ on two levels. On the one hand, this occurs by giving

priority to the introduction of the knowledge economy and innovation. On the other hand, through the principle of project appeals which leaves local actors responsible for project development, drawing up lists of participants and choosing the geographical area. Involvement of local authorities in defining the axes and methods of territorial development went a step further with the implementation of business clusters. These were principally focused both on very small businesses and SMEs, and upon the development of all types of innovation and policies which most closely aligned to the market. At the time of publication, in 2015, France had 120 enterprise clusters.

The policy of competitiveness poles was launched in 2004. Highly inspired by spatial analysis of production gains, it aimed to institute collaboration between businesses and territories and to support R&D projects financially, linking businesses and actors within public research. Indeed, France was reputed for the low number of collaborations between economic actors (big business, SMEs and research and higher education centers), which had had the consequence of a production level which was inferior to its full potential and that was achieved by localized businesses in other comparable countries, and especially in Germany.

The implementation of this policy fell within the bottom-up approach. In response to project appeals, candidates (groups of businesses, higher education establishments, private and public research centers, and local authorities) defined a development strategy starting from a set of themes (a market, scientific and technological sphere), a list of potential partners, medium-term objectives and finally by specifying the project territory area (firstly at a commune level in 2005 then during subsequent waves at a regional level).

The various dimensions of this policy are reflected by the multiplicity of bodies responsible for its implementation. It brings together the public bodies in charge of business support programs (the Ministry for Economy, Industry and Employment, Oséo/BPIFrance, *la Caisse des Dépôts et Consignations*⁸, public research support institutions (e.g. MESR⁹, *Agence Nationale de la Recherche*¹⁰), and those specializing in regional planning

8 The latter three organizations providing finance.

9 The Ministry of Higher Education and Research.

10 The National Research Agency.

(DATAR¹¹, the Ministry for the Overseas Territories). The involvement of local authorities in the operation and financing of the poles in their respective territory allows for a territorial anchorage mechanism. Public funds were used to achieve this objective. The *Fonds Unique Interministériel*¹², was financed by the state with specific programs managed by local authorities.

During its effective implementation, this policy was subject to conflicting pressures, with the addition of the regional planning objective to the original productivity and competition objectives. The profusion of poles (71 ratified in three waves from July 2005 to July 2007), the creation of several poles in the same field (ten within the food processing sector and five in the transport sector) and also their distribution across the entire territory was subject to criticism. Critics used the argument that these choices were gutting the objectives of the policy.

Box 2.3. Competitiveness poles in France

There are various types of entrepreneurial ecosystems. They can be very specialized. For example, this is the case for *Céréales Vallée* (or Cereals Valley) in the Clermont-Ferrand area or Plastipolis in Oyonax, which specializes in plastics processing. They may also group various industries working in the same field together, such as Moveo Automobiles in Rouen or Aerospace Valley near Toulouse. They are geographically defined but not systematically associated with a campus, a town, or a *département*. The most symbolic case of multi-localization is Cap Energie whose main operations are located at Saint-Paul-lez-Durance but which includes operations in Corsica and Guadeloupe. Cosmestic Valley is equally interesting, being a competitiveness pole which is situated in two French regions – Centre and Val de Loire. The map shown in Figure 2.5 takes account of the plurality of French competitiveness poles also indicating the various categories of poles instituted (that is to say worldwide, world market and regional).

11 *Délégation Interministérielle à l'Aménagement du Territoire à l'Attractivité Régionale* with responsibility for regional planning in France.

12 The so-called single inter-ministry fund.

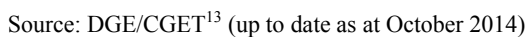


Figure 2.5. Map of competitiveness poles

13 *Direction générale des Entreprises* (i.e. Directorate General for Enterprise and Industry) and *Commissariat Général à l'Égalité des Territoires* (i.e. the Commission for Territorial Equality).

2.3. Strategies and public policies favoring entrepreneurial ecosystems

2.3.1. *Definite risks of business creation*

From the previous sections, it is clear that the concept of territory cannot be reduced to a collection of resources (whether material, natural or human, etc.) developed by businesses. It also involves a created space, into which economic and social relationships (which may be both formal and informal) are embedded. The latter fall within the history and sociology of territories (see section 2.1).

This distinctive feature calls for appropriate policies. Public decision-makers are confronted with the challenge that currently, even within environments which favor innovation, the number of high-growth businesses remains low. There were only three releases of the so-called *Plan Gazelle*, which was launched in 2007 by the French government. The research carried out at INSEE¹⁴ by Claude Picart demonstrated that high-growth businesses were not in a position to maintain this rate for more than a few years and that this short-term over-performance was achieved at the price of declining prospects in the medium- to long-term.

This separation of increased rates of business creation and a level of innovation, which has reached a ceiling, is not unique to France. It has been noted in other European countries as a recent OECD report illustrates¹⁵. The high percentage of so-called entrepreneurs “out of necessity”¹⁶, to use the expression of the Global Entrepreneurship Monitor, might explain this difficulty of transforming new businesses into high-growth businesses. It is generally appreciated that environments characterized by a high rate of start-up are not necessarily those which are the most dynamic or the most resilient

14 *Institut National de la Statistique et les Études Économiques*, i.e. French National Institute of Statistics and Economic Studies.

15 This is available at the following link: <http://www.oecd.org/cfe/leed/Entrepreneurial-ecosystems.pdf>.

16 So-called French “auto-entrepreneurs” (sole traders) are representative of this situation.

when faced with economic crises. The reason is found in the nature of start-ups.

An examination into start-up entrepreneurs demonstrates that France, as with the majority of other OECD countries, is characterized by a preeminence of sole traders, who are essentially motivated by becoming self-employed (being entrepreneurs “out of necessity”). The results of INSEE’s SINE survey¹⁷ show the important role of unemployed start-up entrepreneurs and the non-active population amongst new entrepreneurs. The creation of auto-entrepreneur status in 2009 and the vast range of start-up support for those in financial difficulty (ACCRES¹⁸, micro-credit and loans on trust, etc.) favors the creation of new businesses, and to a certain extent, improves their survival prospects [DES 11]. However, these have little effect on national or territorial competitiveness. It could be considered that the more significant the number of businesses created, the greater the probability of finding innovative high-growth enterprises. Several empirical studies have shown that this is not the case [STA 13].

In fact, studies of innovative entrepreneurial ecosystems show that high-growth businesses need specific environments to emerge and expand; and that the support mechanisms for business creation do not favor them [MOT 12]. The operation of innovative entrepreneurial ecosystems calls for the mobilization of specific resources, the majority of which bears no relation to the mechanisms intended for business creation.

This separation between business start-ups and the creation of entrepreneurial ecosystems has important consequences for the implementation of public policies favoring territorial competitiveness. Firstly, the efforts to stimulate competitive entrepreneurship cannot be reduced to top-down initiatives envisaging the creation of institutional frameworks and the infrastructures necessary for welcoming new business in. Both policies and strategies which emanate from the

17 *Système d’Information sur les Nouvelles Entreprises* – the data-gathering system for new businesses.

18 This is support for jobseekers creating or taking on a business.

territory imply local government and local economic players are also necessary. Secondly, policies favoring entrepreneurial ecosystems must be devised independently of those favoring entrepreneurship. Placing the emphasis solely upon business creation proves of little or no effect as, among the start-ups created, few survive and succeed in creating synergies with other economic players, from which wealth for the local economic milieu results.

2.3.2. A wide range of state assistance

This rupture with regular income from local development has its origins in the creation of large communal amenities and with the arrival of new business, attracted by significant fiscal and social security exemptions, with various forms of assistance. Such “pull” factors are forced upon both decision-makers and those benefiting from public action. Significant paradigm shifts result from both with regards to public action and is representative of the economy.

2.3.2.1. Assistance to businesses or assistance with relationships between economic players

In the context of budgetary restrictions and control of public expenditure, the question of business assistance merits consideration. Available evaluations and reports, especially those from the *Cour des Comptes*¹⁹, emphasize the low level effectiveness of assistance and direct subsidies and the opportunities that they bring about. It is, however, striking to note that, first and foremost, actual public expenditure remains focused upon this form of individual action. This choice is all the more surprising given that, while political discourse uses the cluster notion in abundance, the presence of the entrepreneurial ecosystem and clusters has never been stronger.

Although the analysis of this hiatus, not to say contradictions, between the predictions and their translation into financial flows falls within political or administrative science, evidencing the results to expect from these different means of action may contribute to achieving successful negotiations. The construction of the typology

19 The French equivalent of the US Government Accountability Office.

suggested here cuts across the means of action (for example, subsidies or fiscal and social security exemptions) often raised by the proponents of the efficiency-based approach, to two other political dimensions which have been implemented. On the one hand, it is a matter of their nature defined by the intended collective objective (for example, employment support or innovation) and on the other hand, of both their conception and completion (for example, by the state, local authority, or consular bodies at the politico-administrative level). This partition, which is built around two major dividing lines²⁰, brings to light the diversity of alternative models of incentive policies from an economic point of view.

Policies can primarily be classified as per their objectives. Such policies may have two objectives:

- to transform player behavior to allow them to be more successful in fulfilling known initiatives and those already being managed;
- to encourage them to adopt new behaviors by substituting these with others (computerization or automation of tasks previously completed “by hand”) or by introducing a genuine new concept into the business (for example, penetrating international markets).

2.3.2.2. *Forms of action*

Public policies may also be distinguished according to the forms of action that they galvanize. Alternatively, they may favor distortion of the right to set market prices, as with the French adaptation of the Small Business Act or introducing a conditional reward system, as with the financing granted to the joint ventures of several types of enterprises (between SMEs and big business involving businesses and universities, etc.).

Four families of political incentives result from this dual subdivision. They are summarized by the following grid (Figure 2.6).

²⁰ In its original form, the grid was designed from a chart put forward by Olivier Favereau and Suzanne Quiers-Valette [FAV 98]. Carré and Levratto called it “the complete overview of the typology of suggested economy-supporting public policies” [CAR 09].

Adopting the point of view of business operation and the way in which it is distorted by the mechanisms for public action, two typical cases may be identified:

- relative price distortions improve economic and financial performance owing to individual adaptations which allow them to slot into a more optimum segment on the supply curve;

- incentives shaped by integration into a cooperative which favors insertion into joint ventures and communal programs which allow businesses to bring to bear various forms of proximity (whether spatial, social, organizational, cognitive and institutional) which are behind localized training that is particular to a territory.

		Objectives		
		Modification of current behaviors	Adoption of new behaviors	
Methods	Improvement in profitability owing to relative price distortion.	Q-I Individual growth. Investment assistance. Reduction in social security contributions. Access to public procurement.	Q-II Organizational efficiency. Advisory services. Start-up support. R&D support.	→ Individual decisions
	Incentives conditioned by integration within a collective.	Q-III Pooling resources. Collective action. Sharing costs and indivisibility.	Q-IV Collective competitiveness. Collective research support within competitiveness poles.	→ Collective decisions
		↓ Static efficiency	↓ Dynamic efficiency	

Figure 2.6. *The typology of incentive policies favoring SMEs*

Considered from the point of view of effects produced, two forms of policy effectiveness might occur:

- the reduction of the cost of factor inputs through the effect of subsidies or the pooling of resources, and the potential effects it has

upon volume are all behind an efficiency gain which is described as static. When faced with these new relative input factor prices, the business must adapt its *production process* without modifying it completely. In other words, it positions itself alongside the same production function so as to optimize the existing production combination;

– contrary to this, policies favoring adoption of new behaviors work towards the transformation of internal business organization and relationships between firms. These new practices favor a transition towards a new production function which leads to describing the efficiency which is thus gained as “dynamic”.

2.3.2.3. *From policy ideas to concrete initiatives*

Starting from the typology of public support actions, it is possible to identify four forms (shown in Figure 2.5 as quadrants (Q)) of public action which are intended to improve the operation of the economy through an enterprise initiative:

1) Public actions dedicated to organizational growth (quadrant I). They group together industries seeking to both reduce labor costs and favor investment owing to price variations (exemptions from social security, taxes, etc.) and/or subsidy allocation. Affecting production costs, these initiatives aim to improve both business market position and competitiveness. Exemptions from social security dominate this category and in 2014 amounted to nearly 20 billion euros to which the *Crédit d'Impôt Compétitivité Emploi* (CICE)²¹ and other forms of investment assistance were recently added.

2) Actions favoring organizational improvement (quadrant II) resemble previous ones insofar as they implement price reduction practices. However, the stated objective is of an entirely different nature. It is a matter of creating a new need by changing user behavior. The organization *Fonds Régional de l'Aide au Conseil* (FRAC) forces businesses to repeatedly resort to these services²² by, for example, strengthening their business information and managerial

21 This offers finance to business to increase operational activities and thereby competitiveness.

22 This also offers funds for business set-up and operation.

know-how. Innovation assistance (for example, hiring young doctors) also falls within this particular argument. It is less about decreasing production costs than improving efficiencies directly linked to the growth in factor productivity, which is of relevant consideration.

3) The mechanisms promoting the division of means and thus allowing reduced fixed costs (quadrant III) fall within a collective approach that is often boosted by a public decision-maker. The reduction of operational costs or access to shared public resources assumes the formation of a business collective, often involving SMEs. The margin for maneuver of those SMEs involved can exceed the mere inclusion in the standard framework of “operational costs”, resulting in the adoption of new behaviors.

4) Nowadays, particular attention is given to policies aiming to improve the competitiveness of a production collective (quadrant IV). These measures take the form of a combination of incentives aiming to adopt new behaviors, within the framework of a collective of players and businesses, both large and small, of public institutions, in particular within the field of research. Analytically speaking, this category advances cooperative behaviors which fall within both cooperative and proximity reasoning. These policies aim to anchor cooperative behaviors within specific fields, first and foremost research and innovation, which may, for example, achieve evolutionist approaches.

A two-pronged movement of public action mechanisms has taken shape during the course of the last thirty years. On the one hand, fiscal expenses, which take the form of tax and social security exemptions, are increasingly substituted for direct aid that each business might obtain in isolation. On the other hand, the creation of synergism and collective territorialized dynamics has become a priority in the distribution of public resources from the state (competitiveness poles) and local authorities. This shift towards policies directed upon collective competitiveness gains may be viewed as a transformation of the thrust of positive means of business support. From being so-called “collective consumers”, businesses, like the consumers themselves, have increasingly become co-producers of innovation within the framework of policies for competitiveness poles or their various

manifestations. The observer will see in this transformation a measure of the degree of success of this policy. Nevertheless, one can envisage the difficulties in evaluating this change, or indeed of assessing the impact of this type of policy. It implies a significant change in the practices and behavioral reasoning of the businesses which these policies are striving to fulfill. Their assessment requires an analytical framework and indicators which allow them to be distinguished.